## Unit 3: Equations of Lines and Planes

Complete the following table:

| Type of Equation | Lines in $R^2$ | Lines in $R^3$ | Planes (in $R^3$) |
|---|---|---|---|
| Vector | $\vec{r} = \vec{r_0} + t\vec{m}$, $t \in \mathbb{R}$<br>$\vec{m}$ = direction vector | $\vec{r} = \vec{r_0} + t\vec{m}$, $t \in \mathbb{R}$<br>$\vec{m}$ = direction vector. | $\vec{r} = \vec{r_0} + s\vec{a} + t\vec{b}$, $s,t \in \mathbb{R}$<br>$\vec{a}, \vec{b}$ direction vectors. |
| Parametric | $\left.\begin{aligned} x &= x_0 + ta \\ y &= y_0 + tb \end{aligned}\right\} t \in \mathbb{R}$ | $\left.\begin{aligned} x &= x_0 + ta \\ y &= y_0 + tb \\ z &= z_0 + tc \end{aligned}\right\} t \in \mathbb{R}$ | $\left.\begin{aligned} x &= x_0 + sa_1 + tb_1 \\ y &= y_0 + sa_2 + tb_2 \\ z &= z_0 + sa_3 + tb_3 \end{aligned}\right\} s,t \in \mathbb{R}$ |
| Symmetric | $\frac{x-x_0}{a} = \frac{y-y_0}{b}$ | $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ | |
| | ** can have combination of parametric and symmetric if either $a, b, c = 0$ | | |
| Cartesian (Scalar) | $Ax + By + C = 0$<br>$\vec{n} = (A, B)$ | | $Ax + By + Cz + D = 0$<br>$\vec{n} = (A, B, C)$ |

| | Parallel if: | Perpendicular if: |
|---|---|---|
| Lines are | $\vec{m_1} \nparallel \vec{m_2}$, $\vec{m_1} = k\vec{m_2}$ | $\vec{m_1} \perp \vec{m_2}$, $\vec{m_1} \cdot \vec{m_2} = 0$ |
| Planes are | $\vec{n_1} \nparallel \vec{n_2}$, $\vec{n_1} = k\vec{n_2}$ | $\vec{n_1} \perp \vec{n_2}$, $\vec{n_1} \cdot \vec{n_2} = 0$ |

| | Line in $R^2$ | Line in $R^3$ | Plane in $R^3$ |
|---|---|---|---|
| Minimum Distance from a point to $P_0(x_1, y_1, z_1)$ | $d = \frac{\lvert Ax_1 + By_1 + C \rvert}{\sqrt{A^2 + B^2}}$ | $d = \frac{\lvert \vec{P_0P} \times \vec{m} \rvert}{\lvert \vec{m} \rvert}$ | $d = \frac{\lvert Ax_1 + By_1 + Cz_1 + D \rvert}{\sqrt{A^2 + B^2 + C^2}}$ |

| | Long way to check: | Fast way to check: |
|---|---|---|
| Three vectors in $R^3$, $\vec{u}, \vec{v}, \vec{w}$ are coplanar | Let $a, b \in \mathbb{R}$ s.t.<br>$\vec{w} = a\vec{u} + b\vec{v}$<br>solve for unique $a$ & $b$. | triple scalar product.<br>$\vec{u} \cdot (\vec{v} \times \vec{w}) = 0$<br>if coplanar. |

**Practice Questions**

1. In the vector equation $\vec{r} = (3,2,-1) + t(4,0,-2), t \in R$, a point on the line is _____ and a direction vector for the line is _____.

2. In the parametric equations $x = -3 + 2t, y = 4 + t, z = 2 + 7t, t \in R$, a point on the line is _____, and a direction vector for the line is _____.

3. In the symmetric equations $\frac{x-1}{3} = \frac{2-y}{4} = z + 5$, a point on the line is _____ and a direction vector of the line is _____.

4. The minimum distance of the point $Q(2,-4)$ to the line $2x + 3y - 7 = 0$ is _____.

5. In a vector equation, $\vec{r} = (3,-7,2) + s(-2,4,1) + t(-1,1,1), s, t \in R$, a point on the plane is _____ and two direction vectors for the plane are _____ and _____.

6. Vectors $\vec{u}$ and $\vec{v}$ are a basis for a plane provided they are _____.

7. The plane $2x - 5y + 3z - 4 = 0$ has a normal vector $\vec{n} =$ _____.

8. If $\vec{u}, \vec{v}$, and $\vec{w}$ are three vectors lying in the same plane and if $\vec{u}$ and $\vec{v}$ are non- collinear, then $\vec{w} =$ _____. These vectors are said to be _____.

9. If $P(3,-7,4), Q(1,3,6)$ and $R(-1,2,-5)$ are three points in a plane, then two vectors in the plane are _____ and _____.

10. To find the normal to the plane in #9, you should find _____.

11. The Cartesian equation of the plane $\vec{r} = (-2,3,2) + s(1,-3,0) + t(-3,1,1), s, t \in R$ is _____.

12. The planes $2x - 5y + 4z + 12 = 0$ and $4x - 10y + 8z = 1$ are parallel because _____.

13. To quickly determine if three vectors form a basis for space, what calculation should be performed? _____.

**ANSWERS:**

1. point: (3,2,−1); direction vector: (4,0,−2) (or any scalar multiple!)
2. point: (−3,4,2); direction vector: (2,1,7) (or any scalar multiple!)
3. point: (1,−2,−5); direction vector: (3,−4,1) (or any scalar multiple!)
4. $\frac{15\sqrt{13}}{13}$ units
5. point: (3,−7,2); direction vectors: (−2,4,1), (−1,1,1) (or any scalar multiples of those!)
6. not collinear (i.e. linearly independent)
7. (2,−5,3)
8. $\vec{w} = a\vec{u} + b\vec{v}$; coplanar
9. (−2,10,2) or (2,1,11) or (4,−9,9) (many answers possible - create a vector between any pair of points and/or use scalar multiples of any of those vectors created)
10. the cross product of any two non-collinear direction vectors
11. $3x + y + 8z - 13 = 0$
12. their normal vectors are scalar multiples of one another
13. the triple scalar product